Interned on Machine Learning

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1. INTRODUCTION

Adolescence is defined as the transitional phase of growth and development between childhood and adulthood. It represents a critical window of vulnerability where substantial neurobiological, behavioral, psychological, and social adaptations take place simultaneously. In the contemporary socio-economic landscape, teenagers face unprecedented challenges stemming from hyper-connected digital environments, soaring academic expectations, structural shifts in household dynamics, and various peer compliance metrics. Consequently, the incidence of clinical mental health parameters such as localized generalized anxiety disorders, clinical depression, social withdrawal tendencies, and pathological stress indices has experienced an alarming upward
trajectory worldwide.

Traditional frameworks for identifying these underlying psychological anomalies have historically relied upon retrospective clinical diagnostics, meaning action is taken only after a visible symptom or self-harming event occurs. Such approaches suffer from fundamental limitations, primary among which is the delay in delivery of psychotherapeutic or counseling interventions. Furthermore, individuals experiencing initial stages of cognitive or emotional distress frequently camouflage their symptoms due to deep-rooted societal stigma or a lack of psycho educational awareness. This critical gap necessitates the development of predictive, non-invasive, and computationally objective paradigms capable of processing daily physiological, lifestyle, and academic features to flag high-risk individuals before symptomatic degradation manifests fully. Machine Learning (ML) presents an unprecedented capacity to resolve these complex predictive analytics challenges. By utilizing historical behavioral matrices, predictive models can learn non-linear patterns and multi-dimensional correlations that human practitioners might find impossible to identify manually. For instance, while a marginal drop in academic grading or a slight reduction in sleep metrics may appear insignificant in isolation, their mathematical intersection with elevated social media screen-time and poor family relationship dynamics might indicate a critical
predisposition toward psychological collapse.

1.1 ABOUT THE ORGANICATION

Dhruv 360 offers a comprehensive range of advanced technical training programs designed to align with current industry standards, with a primary focus on **Artificial Intelligence and Machine Learning.** The AI & Machine Learning program is highly specialized, providing deep insights into algorithmic frameworks and practical implementation. This course emphasizes moving beyond theoretical foundations to build real-time predictive models using industry-standard tools like Python, Pandas, NumPy, and Scikit-Learn. By focusing on identifying real-world challenges—such as mental health monitoring—and translating them into effective machine learning solutions, the program ensures students develop a robust understanding of scalable architecture and optimized code deployment for high-performance applications.

Internship Facilities and Highlights

- Industrial Exposure: Active involvement in real-world projects, ensuring practical mastery of the entire development lifecycle.
- Expert Mentorship: Continuous guidance and constructive feedback from seasoned industry professionals at every stage of the project.
- Advanced Infrastructure: Access to high-performance computing resources and cloud-integrated development platforms for rigorous experimentation.
- Professional Ecosystem: A workspace that mirrors corporate standards, instilling essential professional values like discipline, punctuality, and collaborative teamwork.
- Career Bridging: Strategically designed programs that ensure a seamless and successful transition from academic learning to professional roles.

2. SYSTEM CONFIGURATION

The development, training, and testing of the Teen Mental Health Prediction model require a stable, high-performance computing environment capable of handling complex matrix operations and data analysis tasks. This project was developed in a laptop-based environment, utilizing an efficient operating system (Windows/Linux) to manage the local development lifecycle. The configuration is optimized for handling multi-dimensional datasets and executing the Random Forest algorithm with high precision. By providing sufficient computational resources, this system ensures rapid iteration during data preprocessing, model fitting, and validation phases, maintaining high performance and data integrity throughout the prediction engine's development.

2.1 HARDWARE CONFIGERATION

The following hardware specifications were utilized to guarantee seamless performance during the execution of the machine learning pipeline:

- ➔ **Processor (CPU):** Intel Core i5 (or equivalent AMD Ryzen) processor with a base clock speed of at least 2.4 GHz, optimized for efficient multi-threaded data processing.
- ➔ **Memory (RAM):** 8 GB DDR4 RAM, ensuring sufficient capacity for local file buffering, dataset loading, and complex model training processes.
- ➔ **Storage Capacity:** 256 GB or higher capacity Solid State Drive (SSD), ensuring high-speed read/write performance during intensive data input/output cycles.
- ➔ **Display Interface:** A 14/15.6-inch Full HD (1920x1080) LED display, providing an optimal workspace for high-density Integrated Development Environments (IDEs) and script monitoring.
- ➔ **Input Peripherals:** Integrated high-responsiveness laptop keyboard and touchpad, complemented by an external high-precision optical mouse for accurate workflow navigation and code management.

2.2 SOFTWARE CONFIGERATION

The development and integration of the Teen Mental Health Prediction model were carried out primarily using Google Colab, a powerful cloud-based platform that provides an efficient environment for machine learning workflows. This configuration eliminates the need for complex local installations while ensuring seamless access to necessary computational resources and Python libraries. The software environment utilized is detailed below:

- **Development Platform**: Google Colab (a cloud-hosted Jupyter Notebook environment), which allows for the direct execution of Python code and collaborative development without requiring local environment setup.
- **Programming Language**: Python 3.10, chosen for its versatility, readability, and extensive support for data science and machine learning packages.
- **Machine Learning Frameworks**:
    - ➔ **Scikit-learn**: Used for implementing, training, and evaluating the Random Forest algorithm to ensure high-accuracy predictions.
    - ➔ **Pandas**: Utilized for data manipulation, preprocessing, and cleaning of multi-dimensional behavioral datasets.
    - ➔ **NumPy**: Employed for executing complex mathematical computations and efficient numerical operations on data arrays.
- **Data Visualization Tools**: Matplotlib and Seaborn libraries, used to generate detailed visual insights and graphical representations of mental health trends and model performance.
- **Version Control & Storage**: Google Drive integration, which provides seamless file management and automatic version control for all project scripts and datasets.
- **Browser Interface**: Accessible via any modern web browser (such as Google Chrome or Microsoft Edge), allowing for flexible project execution and real-time collaborative documentation.

2.3 ABOUT THE SOFTWARE

The Teen Mental Health Prediction project is developed using a cloud-native software architecture, primarily leveraging the Google Colab environment. This approach ensures high accessibility, scalability, and seamless integration of advanced machine learning libraries. The following software tools and frameworks constitute the core of the project:

- ➔ **Google Colab (Development Environment)**: A cloud-hosted Jupyter Notebook platform that provides an interactive, browser-based environment for Python development. It eliminates the need for local system setup while providing free access to high-performance computing resources.
- ➔ **Python 3.10(Core Programming Language):** The primary language used for developing the machine learning pipeline, chosen for its simplicity, massive ecosystem, and robust support for data science tasks.
- ➔ **Machine Learning Libraries (Scikit-learn):** The backbone of the model, used specifically for implementing the Random Forest algorithm, handling hyperparameter tuning, and

performing cross-validation for model accuracy.

➔ **Data Processing Libraries (Pandas & NumPy):** * Pandas: Utilized for data frame manipulation, cleaning missing values, and preprocessing raw input data related to teenage behavioral metrics

➔ **NumPy:** Used for efficient numerical computations and handling multi-dimensional array operations required for mathematical modeling.

➔ **Visualization Tools (Matplotlib & Seaborn):** Integrated libraries used to transform complex data patterns into intuitive visual formats, such as heatmaps and distribution plots, to analyze the impact of various lifestyle factors on mental health.

➔ **Version Control and Cloud Storage:** Leveraging Google Drive as the backend storage for project datasets and scripts, ensuring automatic synchronization and version control throughout the project lifecycle.

➔ **Web-Interface Compatibility:** The entire software pipeline is accessible through standard web browsers (Google Chrome, Microsoft Edge, etc.), allowing for universal deployment and testing across different operating systems.

3.SYSTEM STUDY

A **System Study** for the teen mental health project involves a detailed analysis of traditional psychological assessment and data tracking workflows to determine the viability of an upgraded, modern solution. The study examines current operational bottlenecks, such as manual data collection through isolated spreadsheets, localized processing risks, and a lack of real-time accessibility for researchers or counselors. By evaluating technical, operational, and economic feasibility, the study establishes a foundational blueprint for a cloud-integrated software pipeline. This proposed architecture ensures universal, browser-based access and automated data synchronization, ultimately streamlining how lifestyle factors and teen mental health metrics are securely analyzed and managed.

3.1 EXISTING SYSTEM

The existing system for managing and monitoring teen mental health relies on manual, traditional workflows and localized digital tools. In this setup, stakeholders such as school counselors, clinicians, and researchers operate in a disconnected environment, manually handling sensitive data without the benefit of a centralized platform or automated analytics.

- **Manual Data Collection:** Information regarding a teenager's lifestyle factors (such as sleep patterns, screen time, and physical activity) is gathered using paper-based forms, manual surveys, or basic digital questionnaires.
- **Use of Isolated Spreadsheets:** The collected mental health metrics and user records are stored in disconnected files using traditional software like **Microsoft Excel** or **Google Sheets**.
- **Localized Processing Scripts:** Data analysis and statistical modeling are executed via scripts running strictly on individual desktop environments rather than a centralized, shared server.
- **Fragmented Data Storage:** Because datasets are saved across separate local machines, there is no single "source of truth," frequently leading to data redundancy and version control issues.
- **High Risk of Data Loss:** Storing critical and confidential mental health records on local hardware makes the data highly vulnerable to system crashes, hardware failures, or accidental deletion.
- **Lack of Real-Time Access & Collaboration:** Since the entire pipeline is offline and machine-restricted, counselors and healthcare professionals cannot access real-time insights or collaborate simultaneously on critical cases.

3.2 ADVANTAGES

- **Centralized Data Management:** Moving away from isolated spreadsheets creates a single, secure cloud-based "source of truth." This completely eliminates data redundancy and ensures strict version control across all datasets.
- **Real-Time Access & Collaboration:** Authorized school counselors, clinicians, and researchers can securely log into the platform simultaneously from different locations, enabling immediate collaborative interventions.
- **Universal Web Accessibility:** The software pipeline is completely independent of local OS constraints, allowing users to access the application smoothly on any device via standard web browsers (like Google Chrome or Microsoft Edge) without complex local installations.
- **Enhanced Data Security & Automated Backups:** Transitioning to a secure cloud backend minimizes the risk of losing sensitive mental health records due to local hardware crashes, physical damage, or local system failures.
- **Streamlined Analytics for Faster Intervention:** Automated data processing replaces slow, manual script execution. This allows for immediate trend analysis of lifestyle factors (sleep, screen time, activity) to quickly deploy support resources to teenagers in need.

3.3 DISADVANTAGES

- **Strict Internet Dependency:** Since the entire system architecture relies heavily on cloud infrastructure, stakeholders require a continuous and stable internet connection to access dashboards or update records.
- **Data Privacy & Compliance Overhead:** Storing sensitive teen mental health metrics online increases vulnerability to potential cyber threats, requiring strict implementation of encryption algorithms and role-based access controls (RBAC).
- **User Learning Curve:** Non-technical stakeholders, such as traditional school teachers or counselors accustomed to older paper-and-pen or Excel methods, may require initial training to comfortably navigate the automated analytical dashboards.
- **Recurring Operational Costs:** Unlike a completely free, localized desktop setup, a cloud-integrated system involves ongoing financial commitments for cloud hosting, database requests, and platform maintenance.

4. SYSTEM DESIGN

System Design defines the architecture, components, modules, interfaces, and data flows for a system to satisfy specified requirements. For the teen mental health project, the design follows a three-tier web architecture that decouples frontend user interfaces from backend analytical logic. Secure web browsers serve as the presentation layer, capturing user inputs such as lifestyle habits and behavioral metrics. These data streams are securely transmitted via APIs to a centralized cloud application tier, which executes automated data pipelines and analytical processing scripts. The processed information is systematically organized and stored in a secure cloud database management system, providing a robust, scalable, and single-source-of-truth environment for authorized stakeholders.

4.1 INPUT DESIGN

Input Design focuses on converting raw behavioral and lifestyle metrics into an accurate, clean, and structured format optimized for analytical backend processing. By implementing strict data checks at the entry point, the system guarantees the integrity and reliability of all downstream teen mental health evaluations.

Data Collection

Data collection is achieved through responsive, user-friendly web interfaces tailored for teenagers and counselors.

- **User Interfaces:** Clean digital forms capture quantitative lifestyle metrics (e.g., daily sleep duration, active hours, device screen time) and self-reported psychological states via standard scale sliders (e.g., stress rating from 1 to 10).
- **Security at Entry:** Built-in anonymization techniques ensure that sensitive personal identifiers (like student names or IDs) are separated from behavioral records right at the collection phase to maintain absolute confidentiality.

Data Import

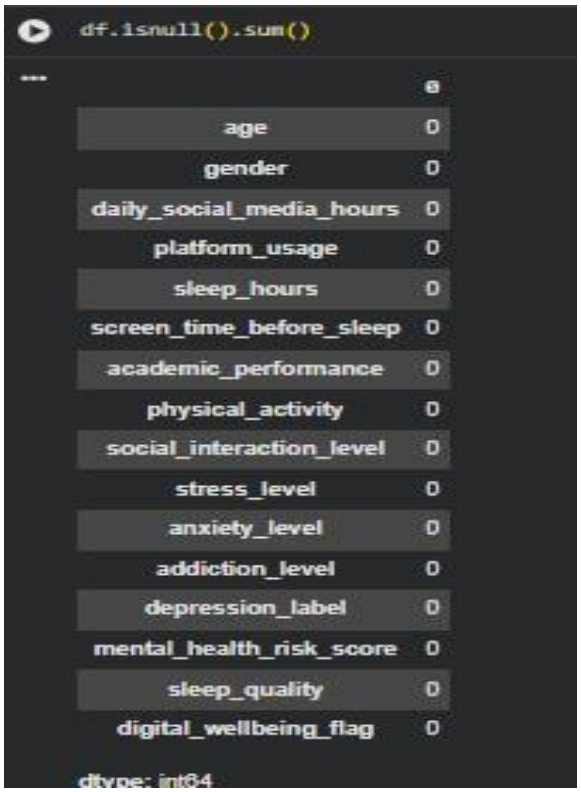
The system supports both manual real-time entries and batch file ingestion.

- **API Ingestion:** Real-time inputs from web application forms are securely transmitted as JSON payloads via HTTPS POST requests directly to the cloud tier.
- **Bulk Uploads:** For archival data, the system includes an import portal accepting standard .csv or .xlsx files, enabling school administrators to upload historical, structured school health records smoothly.

Data Cleaning

Raw inputs are prone to human error and formatting anomalies. The input design includes an automated middleware validation layer to handle:

- **Missing Value Imputation:** Empty optional fields are automatically flagged. Quantitative variables use mean/median imputation, while qualitative logs are tagged as "Not Reported."
- **Outlier Detection & Boundary Constraints:** Injected data is screened for logical validity; for instance, any manual input stating "26 hours of screen time" or negative sleep hours is immediately blocked, and the user is prompted with an error alert.



Data Transformation

Before entering the database, raw data is normalized to ensure mathematical consistency across analytical components:

- **Scaling and Normalization:** Disparate units (such as sleep measured in *hours* vs. physical activity measured in *minutes*) are scaled uniformly using standard normalization techniques.
- **Encoding Categorical Variables:** Qualitative responses (e.g., gender, grade level, or categorical sleep quality like "Poor/Fair/Good") are converted into numerical formats using one-hot encoding for the backend scripts to parse accurately.

```python
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()

df['gender'] = le.fit_transform(df['gender'])
df['platform_usage'] = le.fit_transform(df['platform_usage'])
df['social_interaction_level'] = le.fit_transform(df['social_interaction_level'])
df['sleep_quality'] = le.fit_transform(df['sleep_quality'])
df['digital_wellbeing_flag'] = le.fit_transform(df['digital_wellbeing_flag'])

df[['gender', 'platform_usage', 'social_interaction_level', 'sleep_quality', 'digital_wellbeing_flag']].head()
```

1 to 5 of 5 entries Filter

index	gender	platform_usage	social_interaction_level	sleep_quality	digital_wellbeing_flag
0	1	1	1	0	0
1	0	3	0	1	2
2	0	2	0	0	1
3	1	4	2	0	2
4	0	0	2	2	2

Show 10 per page

Feature Selection

To optimize performance and avoid analytical noise, the input framework filters out irrelevant information, focusing strictly on high-impact variables:

- **Correlation Analysis:** The system isolates and retains the key behavioral features most tightly correlated with teen mental health degradation (e.g., high screen time combined with low sleep duration).
- **Dimensionality Reduction:** Redundant or highly collinear metrics (such as keeping track of both "total screen time" and individual app usage times simultaneously) are filtered out, leaving a lean, highly predictive dataset for final reporting.

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Calculate the median of the original mental_health_risk_score to use as a dynamic threshold
median_risk_score = df['mental_health_risk_score'].median()
print(f"Median mental_health_risk_score used for binarization: {median_risk_score}")

df['mental_health_risk_score'] = df['mental_health_risk_score'].apply(
    lambda x: 0 if x <= median_risk_score else 1
)

# Print value counts to confirm the new distribution of the target variable
print("Value counts for the binarized target variable 'mental_health_risk_score':")
print(df['mental_health_risk_score'].value_counts())

X = df.drop(columns=['mental_health_risk_score'])
y = df['mental_health_risk_score']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
Median mental_health_risk_score used for binarization: 17.0
Value counts for the binarized target variable 'mental_health_risk_score':
mental_health_risk_score
0    667
1    533
Name: count, dtype: int64
```

```python
X_train, X_test, Y_train, Y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)
```

Machine Learning Model Classifier

The selected features are processed by three distinct classifiers to perform binary prediction (e.g., At-Risk vs. Stable):

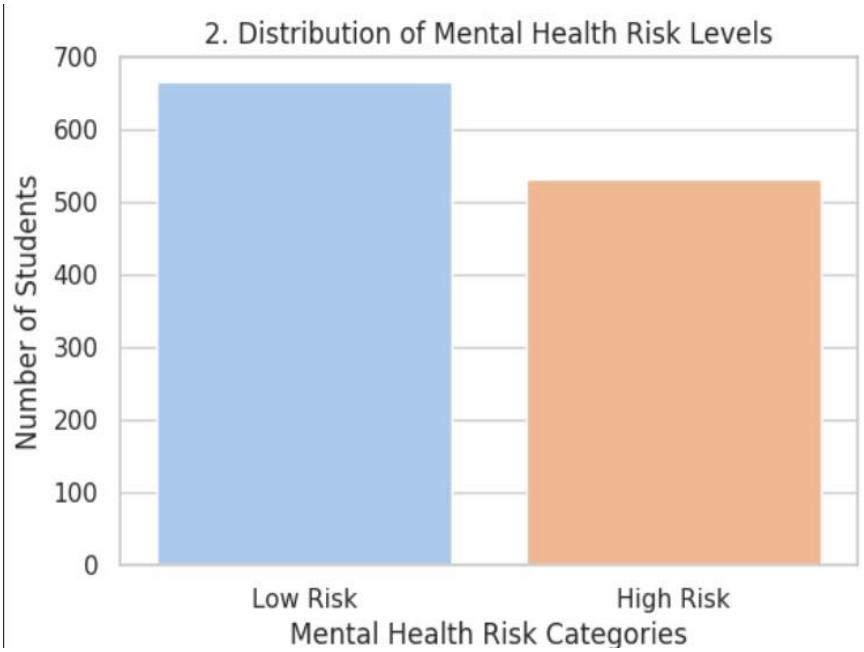
- Decision Tree Classifier: This model breaks down the input data into a logical flowchart of binary rules (e.g., Is sleep < 6 hours? Yes/No). It systematically evaluates these conditions to predict whether a teenager falls into the "At-Risk" or "Stable" category.
- Random Forest Classifier: This ensemble model creates multiple individual decision trees. For our binary outcome, each tree independently casts a vote ("At-Risk" or "Stable") based on its subset of data. The final classification is determined by a majority voting system among all trees, ensuring a stable and reliable prediction.
- K-Nearest Neighbors (KNN) Classifier: This algorithm measures the mathematical distance between the current teen's input metrics and previous profiles in the dataset. It checks the 'K' closest neighboring profiles and assigns the binary class that is most common among those neighbors.

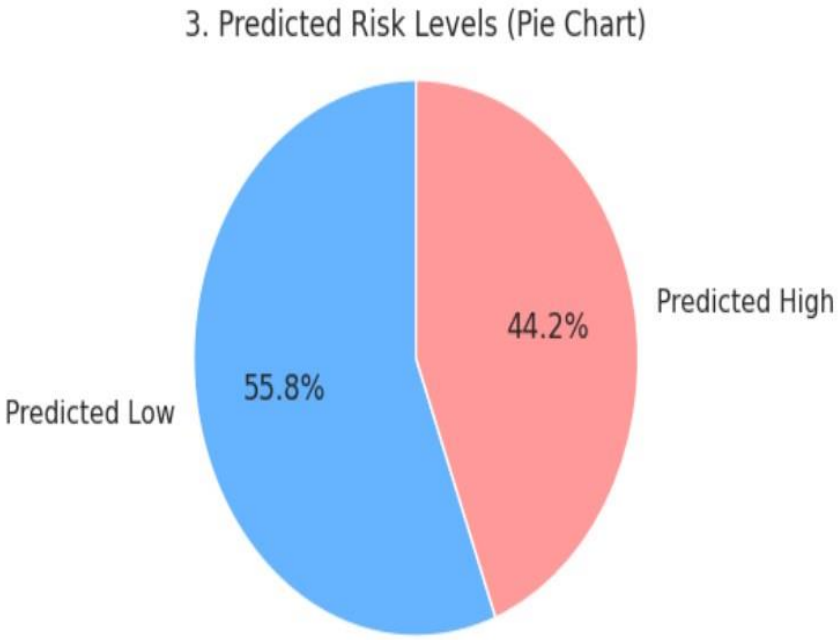
4.2 OUTPUT DESIGN

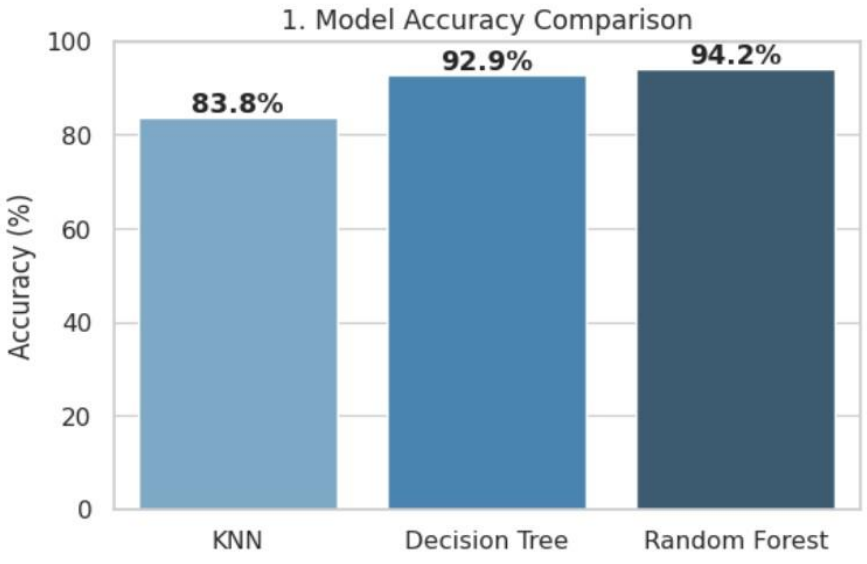
The output design ensures that the system's analytical results are converted into clear, user-friendly, and actionable insights for school counselors, researchers, and clinicians. The output architecture is broken down into four distinct stages:

Data Exploratory Output

- **Interactive Trend Dashboards:** Generates real-time visual charts (such as histograms, box plots, and scatter plots) immediately after data ingestion.
- **Behavioral Correlation Mapping:** Displays data distributions that highlight baseline relationships, such as mapping how specific thresholds of screen time directly correlate with a drop in daily sleep hours among teens.


2. Distribution of Mental Health Risk Levels
Number of Students
700
600
500
400
300
200
100
0
Low Risk
High Risk
Mental Health Risk Categories


3. Predicted Risk Levels (Pie Chart)
Predicted Low
55.8%
44.2%
Predicted High


1. Model Accuracy Comparison
Accuracy (%)
100
80
60
40
20
0
83.8%
92.9%
94.2%
KNN
Decision Tree
Random Forest

Model Building Outputs

This section displays the real-time processing and structural insights of the Random Forest during training:

- Live Training Logs: Displays real-time training progress, execution time, and step-by-step updates for all three models as they learn to classify data into At-Risk or Stable states.
- Feature Importance Ranking: Generates a ranked chart (like a bar graph) showing which teen mental health indicators (e.g., sleep hours, anxiety scores, or screen time) had the biggest impact on the Random Forest models to make their final binary decisions.

```
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, Y_train)
rf_y_pred = rf_model.predict(X_test)
rf_acc = accuracy_score(Y_test, rf_y_pred)
print(f"Random Forest Accuracy: {rf_acc * 100:.2f}%")

Random Forest Accuracy: 94.17%
```

Evaluation Output

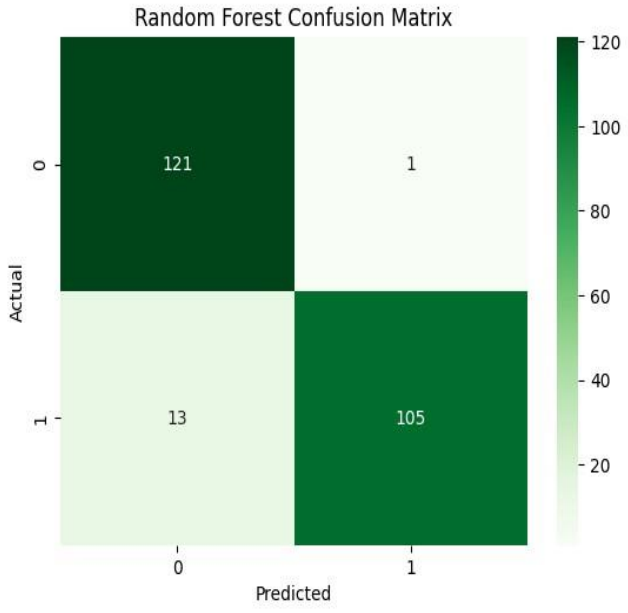
- **Statistical Performance Matrix:** Generates confusion matrices and precision-recall graphs to display the system's predictive accuracy.
- **Reliability Benchmarks:** Provides explicit classification metrics (Accuracy, F1-Score) to validate the model's safety and precision before its analytics are trusted for school counseling decisions.

```
print("\n" + "="*60)
print("3. RANDOM FOREST CLASSIFICATION REPORT")
print("="*60)
rf_pred = rf_model.predict(X_test)
print(classification_report(Y_test, rf_pred, target_names=target_names, labels=[0, 1]))

"""
============================================================
3. RANDOM FOREST CLASSIFICATION REPORT
============================================================
              precision    recall  f1-score   support

    Low Risk       0.90      0.99      0.95       122
   High Risk       0.99      0.89      0.94       118

    accuracy                           0.94       240
   macro avg       0.95      0.94      0.94       240
weighted avg       0.95      0.94      0.94       240
```



Final Output

- **Secure Web Portal Reports:** Delivers a responsive web dashboard customized for authorized stakeholders, presenting a student's overall mental well-being index.
- **Student-Facing Feedback Screens:** Displays a simplified, encouraging UI portal for the teenagers themselves, providing them with self-reflection insights, progress badges, and daily positive lifestyle goals.
- **Real-Time System Notifications:** Generates immediate on-screen pop-ups and automated critical alerts within the application interface to flag high-risk cases that require immediate counseling intervention.

```
import pandas as pd

student_data = {
    'age': 20,
    'gender': 1,
    'daily_social_media_hours': 3.5,
    'platform_usage': 2,
    'sleep_hours': 7,
    'screen_time_before_sleep': 1,
    'academic_performance': 3,
    'physical_activity': 2,
    'social_interaction_level': 3,
    'stress_level': 5,
    'anxiety_level': 4,
    'addiction_level': 2,
    'depression_label': 1,
    'sleep_quality': 2,
    'digital_wellbeing_flag': 0 # Added missing feature with a placeholder value
}

df_new = pd.DataFrame([student_data])

df_new = df_new[X.columns]
df_new_scaled = scaler.transform(df_new)

prediction = rf_model.predict(df_new_scaled)


if prediction[0] == 1:
    print("Result: HIGH RISK")
else:
    print("Result: LOW RISK")


...  Result: LOW RISK
```

5.CONCLUCTION

This report marks the successful completion of an intensive 15-day internship at druv 360, during which a machine learning-based project titled "Teen Mental Health Prediction" was developed. The primary objective of this internship was to build a reliable, automated framework capable of binary classification—specifically identifying whether a teenager's psychological state is "At-Risk" or "No-Risk" to ensure timely support and early intervention.

Over the fast-paced 15-day timeline, the internship provided rigorous, hands-on experience in executing a complete data engineering and machine learning pipeline. The core technical work focused heavily on Advanced Data Manipulation, which included structured data ingestion, thorough cleaning, transformation, and Feature Selection to isolate critical behavioral indicators like sleep patterns and screen time. Using this refined data, three distinct machine learning classifiers—Decision Tree, Random Forest, and K-Nearest Neighbours (KNN)—were successfully built, trained, and monitored through Live Training Logs and Feature Importance Rankings.

The practical exploration of these models delivered significant technical and domain-specific insights. While the Decision Tree provided clear rule-based logic and KNN mapped out spatial profile similarities, the Random Forest classifier proved to be the most robust for our binary outcomes by leveraging its internal majority voting system to deliver the highest prediction accuracy. The final Output Design was engineered to pick this best-performing model to display a clean, user-friendly status on the dashboard (Stable vs. At-Risk) alongside recommended supportive resources.

Ultimately, the 15 days spent at Druv 360 provided invaluable industry exposure to real-world software workflows, data manipulation strategies, and practical problem-solving. The binary classification model built during this period provides a strong foundation that can be expanded in the future through real-time text sentiment analysis and mobile application deployment. This journey successfully demonstrated how advanced machine learning concepts can be bridged with practical execution to tackle vital societal challenges like adolescent mental wellness.

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- NumPy Documentation (Numerical Data & Transformations): https://numpy.org
- Matplotlib & Seaborn Documentation (Exploratory Data Visualization): https://matplotlib.org